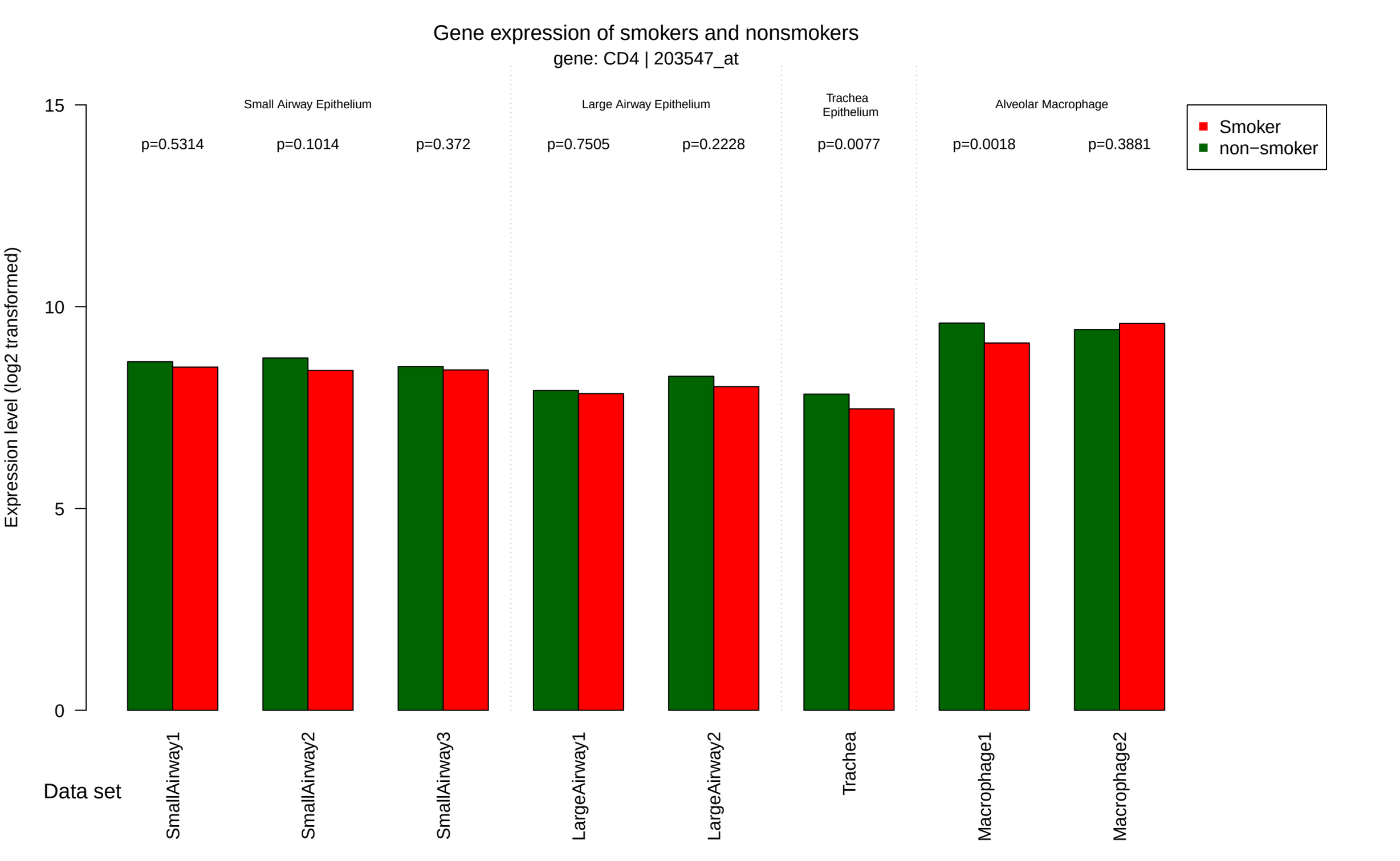
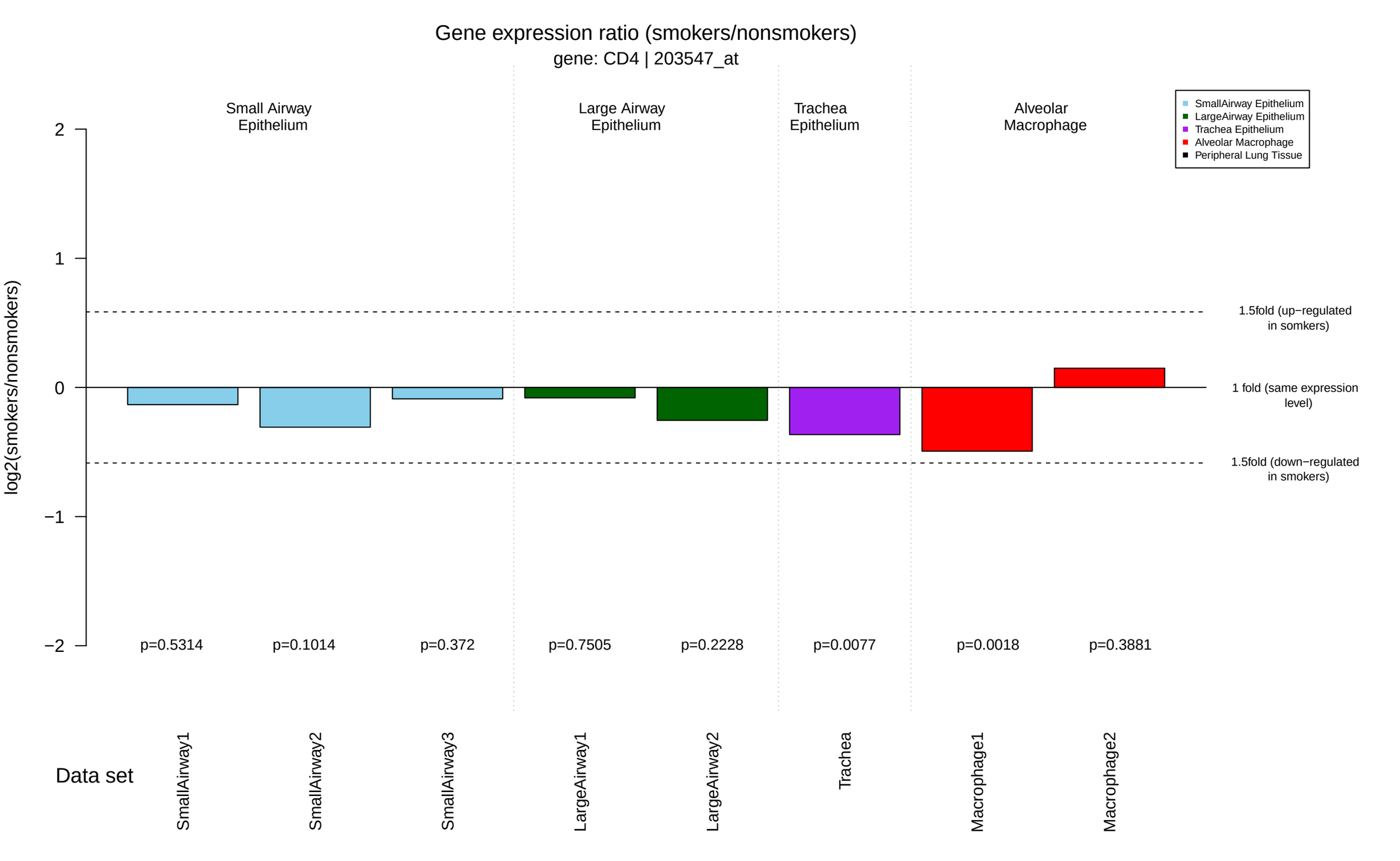
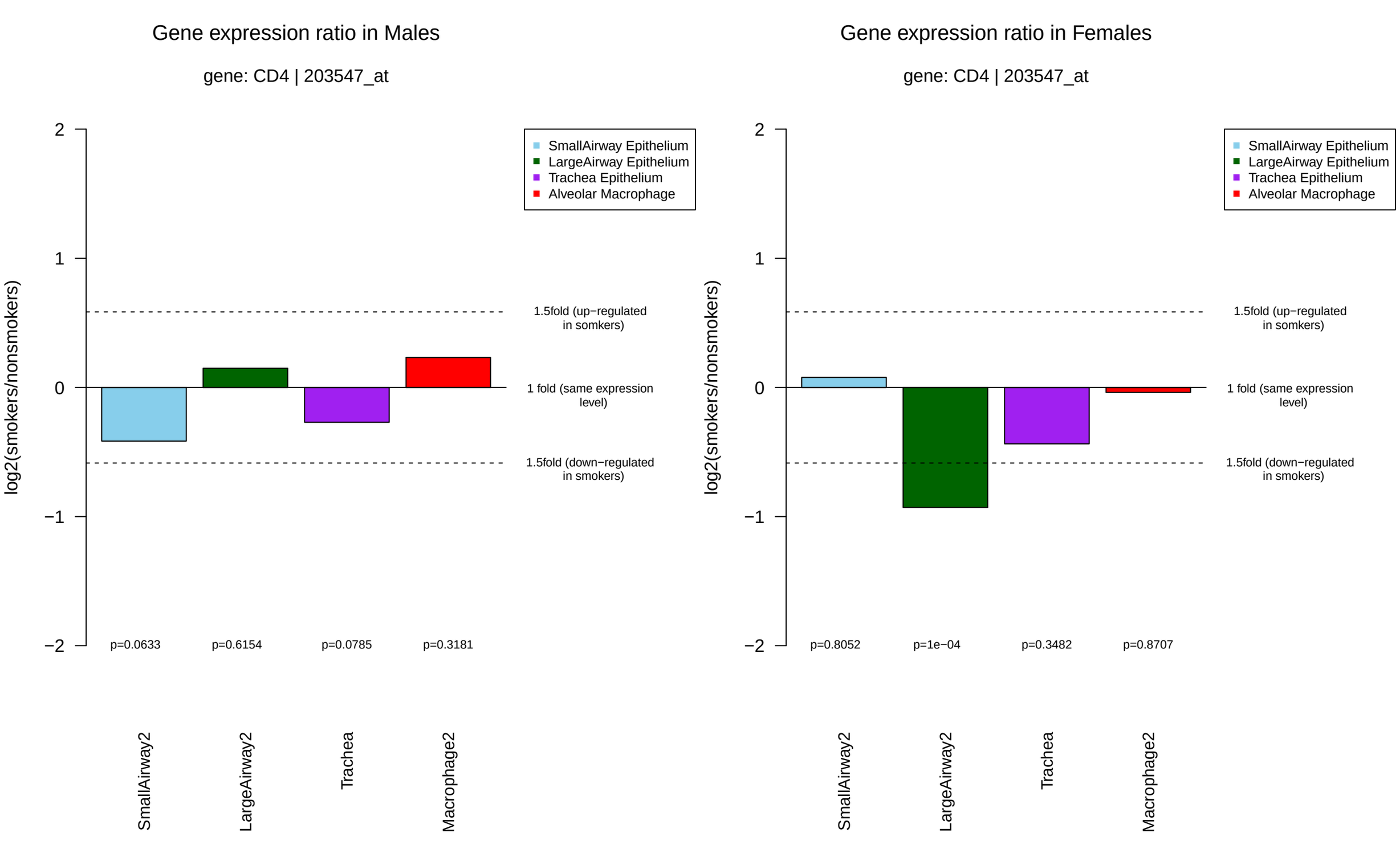
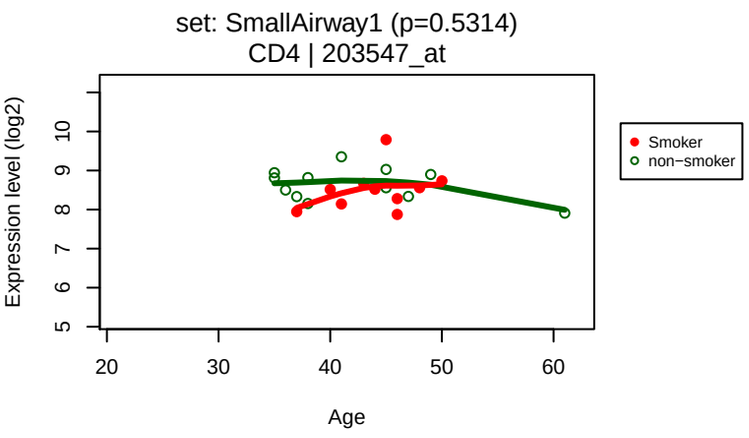
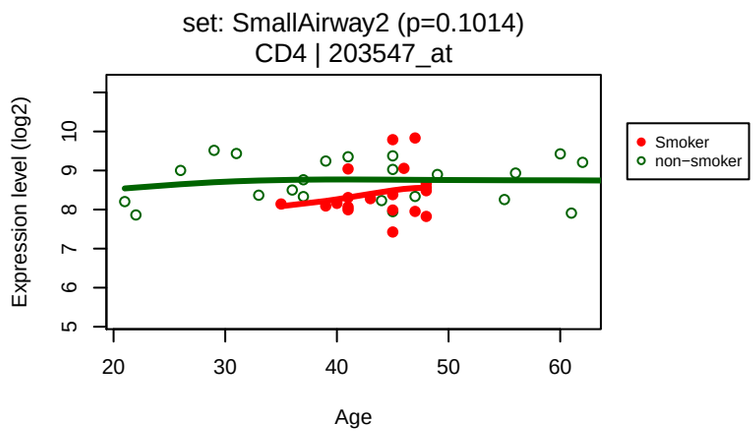
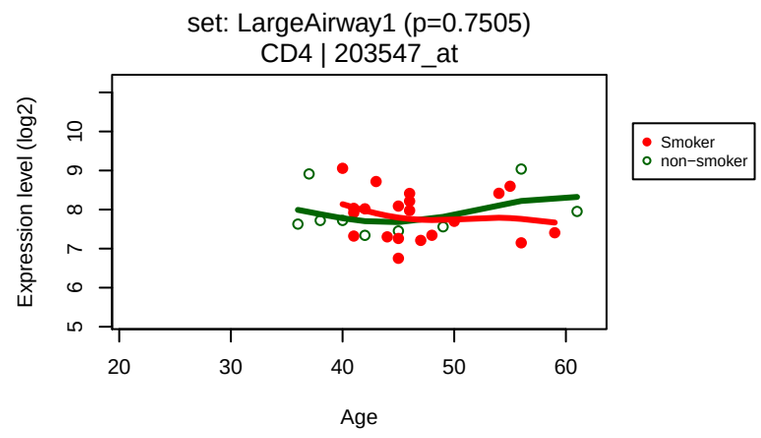

Gene expression of smokers and nonsmokers
gene: CD4 | 203547_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5314
p=0.1014
p=0.372
p=0.7505
p=0.2228
p=0.0077
p=0.0018
p=0.3881
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

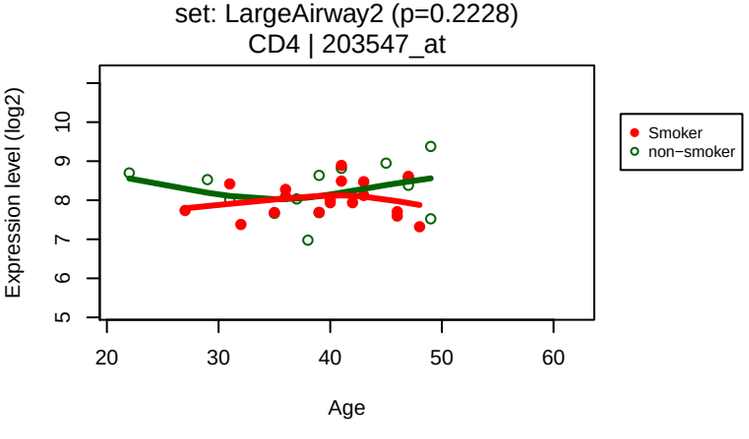

Gene expression ratio (smokers/nonsmokers)
gene: CD4 | 203547_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5314
p=0.1014
p=0.372
p=0.7505
p=0.2228
p=0.0077
p=0.0018
p=0.3881
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

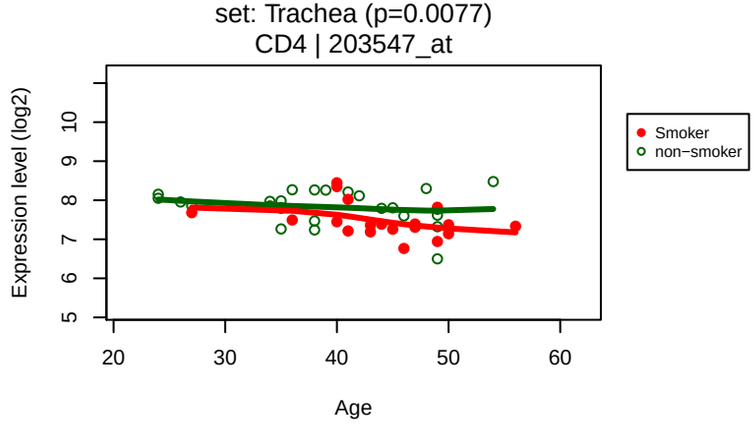

Gene expression ratio in Males
gene: CD4 | 203547_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0633
p=0.6154
p=0.0785
p=0.3181
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CD4 | 203547_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8052
p=1e−04
p=0.3482
p=0.8707
SmallAirway2
LargeAirway2
Trachea
Macrophage2

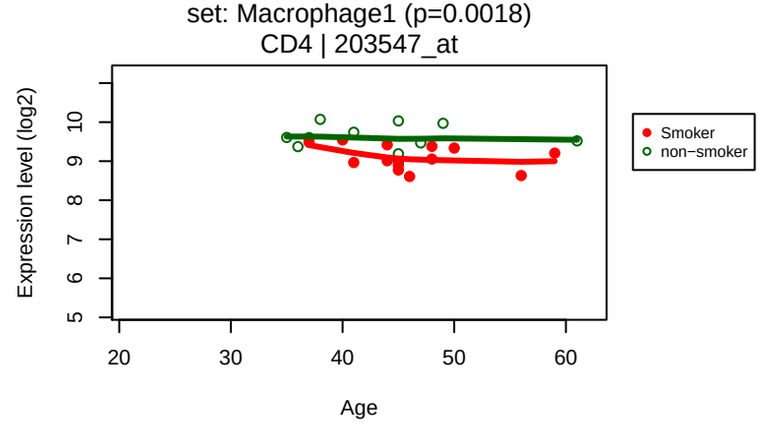

set: SmallAirway1 (p=0.5314)
CD4 | 203547_at
Expression level (log2)
Age
Smoker
non−smoker

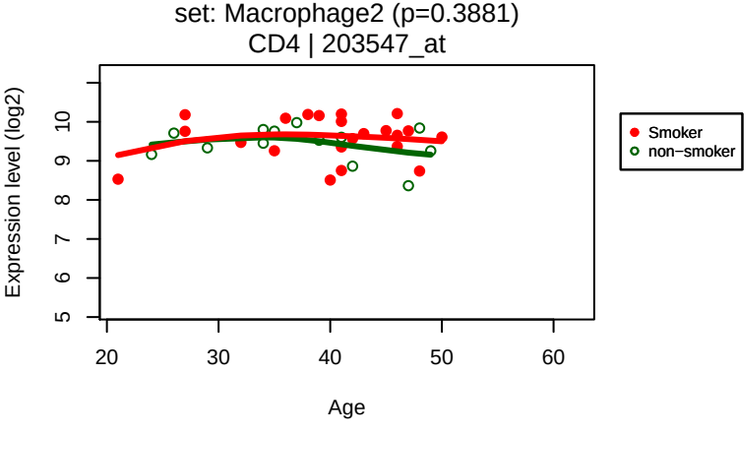

set: SmallAirway2 (p=0.1014)
CD4 | 203547_at
Expression level (log2)
Age
Smoker
non−smoker

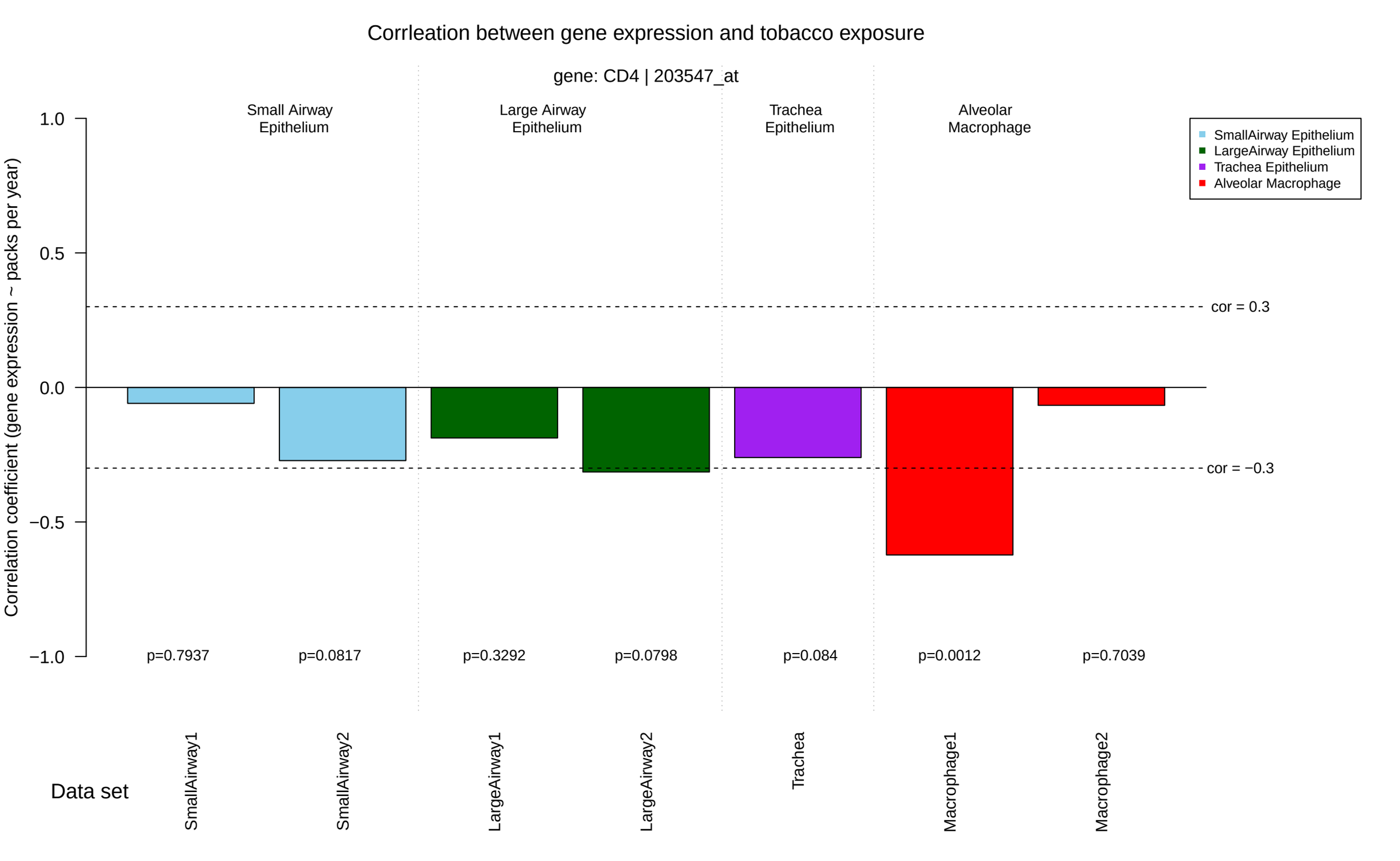

set: LargeAirway1 (p=0.7505)
CD4 | 203547_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2228)
CD4 | 203547_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0077)
CD4 | 203547_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0018)
CD4 | 203547_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3881)
CD4 | 203547_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CD4 | 203547_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7937
p=0.0817
p=0.3292
p=0.0798
p=0.084
p=0.0012
p=0.7039
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2